



Allergen persistence and cross-contamination potential in water reuse within food industry processes

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Food allergies are a critical public health challenge, and many food recalls are due to food allergen detections. Water intensive post-harvest activities are common in the food industry. Thus, the food industry has increasingly treated and reused water within its operations, but these practices are unstandardized and could cross contaminate food products with food allergens. We summarize the evidence on potential food allergens persistence in water reuse in this industry.

Is water reuse in food systems a pathway for allergen cross-contamination?

Water reuse in the food industry. The water–food nexus is becoming a defining challenge for sustainable development since societal growth and food security heavily depend on how water is managed across food production systems^{1,2}. The food industry is increasingly reusing water and treated wastewater effluent to support critical growing, harvesting and processing functions with water treatment targets being tailored to the intended end-use. The use of treated wastewater effluent for crop irrigation is a common practice globally, with regulator-set water quality standards that vary by jurisdiction and often focus on disinfection and microbiological water quality^{3–5}. However, water reuse for water-intensive processes within the food industry is also becoming increasingly common. These water-intensive processes are shown in Fig. 1 and include post-harvest activities, such as: raw product washing (e.g., grain or produce sorting, cutting, washing, chilling); dunk and immersion tanks; dehulling, shelling and peeling tanks; poultry and meat processing; clean-in-place and sanitizing equipment; and produce flumes and conveyors⁶. Many of these activities are essential and intended to remove dirt, debris, and cellular exudates that support microbial growth, thereby mitigating risk from pathogens, spoilage bacteria, chemical contaminants and pesticide residues⁷. Clean-in-place (CIP) operations, necessary to remove fouling and restore equipment performance, can account for 30–80% of operational costs and can consume up to 70% of water, while also requiring substantial chemical use^{8,9}.

Given how volume-intensive and unsustainable these practices are, the food industry has increasingly relied on water reuse and treatment (also known as water conditioning) within its operations to reduce or remove microbiological, chemical, or physical contaminants to allow reuse for one

to more than fifty cycles. By reconditioning the water, the same water can be reused for multiple, different batches of products instead of being disposed of after minimal use⁶. While this is highly desirable from a water conservation perspective, it introduces the potential for cross-contamination of food products with foodborne pathogens and chemical contaminants through contact with contaminated process water¹⁰. Another unexamined aspect of water conditioning in food systems is the potential cross-contamination of cleaning and processing water with food allergens during regular activities, and whether the water treatment trains within food facilities remove these effectively during conditioning. Figure 1 shows water reuse processes, drivers and trends in food systems, how water-intensive these procedures are and how many reuse cycles are often performed. The potential for allergen cross-contamination through food-processing water is influenced by several interacting factors, including the amount of allergenic material entering the water, the extent of water–product contact, frequency of water reuse, persistence of allergenic proteins, effectiveness of water treatment, and opportunities for transfer to subsequently processed products. Processes involving prolonged or intimate contact between food products and recirculated water, such as immersion tanks and produce flumes, may have greater potential for allergen carry-over because allergenic proteins can accumulate in the water and subsequently contact other products. Similarly, processes that involve substantial manipulation and tissue disruption, such as dehulling, shelling, peeling, or cutting, may release greater quantities of soluble and particulate proteins into process water, increasing the potential for subsequent transfer¹¹. The potential for allergen carry-over should therefore be considered a process-level characteristic rather than a direct measure of consumer health risk. A process with high potential for allergen accumulation or transfer does not necessarily result in clinically relevant exposure, as the ultimate risk depends on the concentration and immunoreactivity of the allergen transferred to the finished product, serving size, and the allergen-specific susceptibility of the consumer. In this context, the categories presented in Fig. 1 represent a conceptual ranking based on water–product contact, reuse intensity, allergen loading, protein persistence, treatment performance, and transfer opportunity, rather than a validated quantitative risk assessment.

The nine most clinically relevant food allergens are milk, eggs, peanut, tree nuts, wheat, soybean, fish, crustacean shellfish, and sesame¹². A survey of 50 quality control food industry specialists with knowledge of their company's food allergen management practices showed that 92% of companies produced food products containing one or more of these food allergens¹³. Thus, the potential of allergen persistence in wash or processing water is worth examining. To our knowledge, there has been no research on the occurrence of allergens in food system processing water, and the potential for allergen cross-contamination has not been explored. This review aims to achieve two objectives: 1) summarize relevant knowledge on potential allergens occurrence and persistence in food systems to examine

likelihood of cross-contamination through water reuse; and 2) evaluate the characteristics of allergens that are likely to result in their stability, persistence, fate and transport in food industry water systems.

Relevance of food allergies from a public health perspective

Food allergies are an important public health problem that impacts 1–10% of the global population¹⁴. Food allergy frequency and severity are reported to have increased over recent decades¹⁵. In the United States (US), about 33 million people suffer from food allergies, including 5.6 million children, with childhood food allergies alone resulting in an estimated annual cost of \$24.8 billion^{16–18}. The World Health Organization (WHO) forecasts that nearly 50% of the global population will experience allergies by 2050, making allergies the fourth most common chronic disorder¹⁹.

Food allergies result in an acute Immunoglobulin E (IgE) mediated reaction that occurs in response to specific food components. The allergic reaction is triggered after sensitization occurs as a result of either oral exposure to a food allergen induces an IgE-sensitizing response that primes the host immune system to react, or as a result of maladaptive immune responses in the skin that are now known to also contribute to the development of food allergy¹⁴. Shellfish, fish, tree nuts, and peanuts have been reported to have high rates of allergenic cross-reactivity (Kamath et al.²⁰). Cross-reactivity occurs when structurally similar proteins or allergenic epitopes from different foods are recognized by the same allergen-specific IgE antibodies, potentially resulting in an allergic reaction (Kamath et al.²⁰). However, IgE cross-sensitization does not necessarily result in clinically relevant allergic reactions, and the degree of clinical cross-reactivity varies among food groups and individuals. Allergenic proteins with greater than 70% amino acid sequence identity are generally considered more likely to exhibit IgE cross-reactivity, but cross-reactivity is influenced by the structural similarity and conservation of IgE-binding epitopes and cannot be predicted from sequence identity alone²¹. This implies that food processing operations that reuse water between many different food types could carry over food allergens that trigger some cross-reactivity.

Accidental exposure to food allergens can have major implications. Approximately 7–14% of patients with peanut allergies experience accidental peanut exposure annually²². Similarly, in a prospective 1-year study of 157 adults with physician-diagnosed food allergy, 46% experienced at least one accidental allergic reaction, corresponding to an average of approximately two reactions per affected individual per year; 28% of reported reactions were classified as severe²³. Avoidance remains a key intervention in the management of food allergies²⁴. Several international initiatives attempt to reduce exposure through labeling and testing. Precautionary allergen labeling (PAL) refers to voluntary statements such as “may contain” or “may be present” that communicate the potential unintended presence of allergens resulting from cross-contact during food production. In Australia and New Zealand, the VITAL® (Voluntary Incidental Trace Allergen Labelling) Program provides a standardized, risk-based methodology for assessing allergen cross-contact and determining whether PAL is appropriate. VITAL uses scientifically derived reference doses and action levels to support consistent labeling decisions while reducing unnecessary use of precautionary statements. Still, the presence of undeclared allergens makes compliance difficult with studies in the European Union (EU) estimating that up to 7.6% of “free-from-allergen” products do contain undeclared allergens²⁵. In addition, a study by Soon et al. that reviewed 2,932 food safety incidents between 2008 and 2023 in Canada, the EU, the United Kingdom (UK) and the US, reported the primary reason for food recalls being allergens (28–48%) followed by microbial contamination^{26,27}. The four most commonly reported allergen recalls were for undeclared milk (36%), egg (14%), wheat (13%), and peanut (12%) allergens^{12,28}. To mitigate food recalls, firms

that manufacture products both with and without major allergens need to design proper labels, place the correct product in the correct package, and have a concrete understanding of potential allergen cross-contact pathways during processing²⁹.

Allergen cross-contamination is the unintentional incorporation of a major food allergen in a food during processing. Previous research focused on employee training, cleaning procedures, and managing cross-contamination and found these important for allergen management efforts for companies¹³. Cross-contamination of allergens often results from carryover of allergen-containing material on shared equipment (70% of facilities) or bulk tanks (<40%), which cannot always be prevented especially in small and medium sized facilities²⁶. Very little has been published on the effectiveness of cleaning procedures for removing allergenic materials from processing equipment³⁰. A survey of 50 quality control and allergen management food industry specialists showed that despite 96% of companies having detailed food allergen control plans in place in alignment with best practices, nearly half (42%) had at least one food allergen-related recall within the past 5 years¹³. For food companies, the economic and reputational costs resulting from recalls as a result of mislabeling and cross-contamination are major, which is why companies create programs and practices to ensure that food products with and without major allergens are kept separated in production and distribution²⁹. There are currently no standard water conditioning practices or water reuse guidance in the food industry. There is no data or methods to validate whether allergen cross-contamination through water reuse or ineffective treatment is a concern. Given how recent water reuse in food processing is, it is not surprising that potential of cross-contamination has not been explored for conditioned water.

Allergens' stability and persistence in water systems

Allergen properties in biological systems that result in stability and persistence. Abundance and stability are common features of allergens in biological systems³¹, which have implications for removal or persistence of allergens in water systems. Researchers have proposed that greater abundance of potential allergens may increase the likelihood of human exposure and, consequently, the probability of sensitization³¹. Although allergen abundance may contribute to allergenic risk, sensitization is multifactorial and is influenced by additional characteristics, including protein stability, exposure duration, and immunological factors.

Pekar et al. define allergen stability as the ability of proteins to preserve their native, three-dimensional structure over time after chemical, physical or protease exposure³². Depending on their routes of exposure, food allergens possess molecular and structural properties that can influence their stability, persistence, and ability to survive physiological barriers and enzymatic degradation. For orally ingested allergens, stability to gastrointestinal conditions, including acidic pH and proteolytic and hydrolytic degradation, can allow intact proteins or immunologically relevant peptide fragments to persist and become available for uptake across the intestinal epithelium and subsequent interaction with antigen-presenting cells. Following uptake, allergens undergo intracellular processing and presentation that can contribute to the development of an IgE-mediated immune response. Therefore, allergen stability is one of several factors influencing sensitization rather than a sole determinant of allergenic potential^{31,32}. Respiratory allergens, such as pollen or dust mite proteins, are not exposed to low pH levels or enzymes, but exhibit extensive solubility in the aqueous environments of the mucosal surfaces of the alveoli and have the correct particle size to enter and escape mucus binding³³. Allergens entering the body through the skin must cross the epidermal barrier, where filaggrin and its degradation products

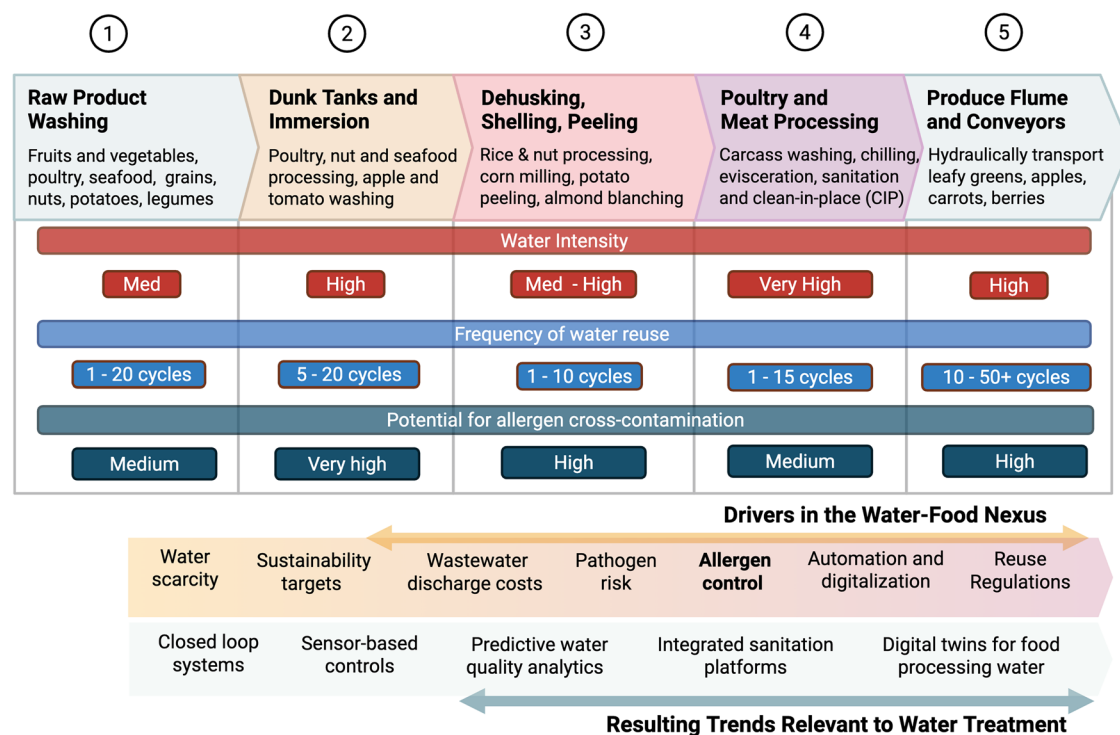


Fig. 1 | Drivers and trends for water reuse in food processing systems and their impacts on allergen cross-contamination potential. Created in BioRender. Maal-Bared, R. (2026).

contribute to the structural integrity and hydration of the stratum corneum, while intercellular lipids form hydrophobic and hydrophilic structures that regulate skin permeability³⁴. Consequently, allergens penetrating intact skin may require physicochemical properties that facilitate passage through the lipid-rich barrier, such as hydrophobicity, or association with a lipophilic carrier. Unlike allergens entering through the gastrointestinal tract, allergens crossing the outer skin barrier are not exposed to the same proteolytic environment and therefore do not require comparable resistance to protein-degrading enzymes³². Furthermore, no specific stability properties are required for intradermally injected allergens (e.g., insect venoms)³³. In this review, we hypothesize that molecular properties that contribute to allergen stability in food and physiological environments may also contribute to the persistence of allergenic proteins during water-based food-processing operations. Such persistence could influence the potential for allergen residues to remain in process water, survive treatment or cleaning processes, and contribute to subsequent cross-contact. This conceptual relationship is illustrated in Fig. 2.

Other factors that result in allergen stability and persistence

While food allergens are diverse proteins, they often do share common structural features, such as strong disulfide bonds or oligomeric structures^{35,36}. Disulfide bonds make proteins more resistance to chemical or biological degradation, influence stability under varying pH and influence protein 3-D structure. Studies showed that peanut allergens Ara h 2 and Ara h 6 are extremely resistant to proteolytic digestion due to their four highly stable disulfide bonds³⁷. Higher abundance of β -sheet structures results in more thermal tolerance³². Thus, disulfide bonds and 3-D protein folding can impact charge, hydrophobicity and protein partitioning in water treatment

processes, which impact whether a protein ultimately partitions into solids or liquid phases.

Food allergens encompass a broad range of molecular sizes, although many are relatively small proteins with molecular masses below approximately 70 kDa³⁸. For example, the major peanut allergen Ara h 1 has a molecular mass of approximately 63–65 kDa, placing it near the upper end of this range³². Studies have suggested that proteins or peptides of approximately 3.5 kDa or greater may retain sufficient structural features to elicit an antibody response, although molecular size alone does not determine allergenicity³². Allergen molecular weight and size range have major implications for removal in water systems, but retention of these molecules on membranes is not solely controlled by size exclusion³⁹. Membrane molecular weight cut-off (MWCO) is defined as the molecular weight of the organic tracer that is 90% retained by the membrane using organic tracer retention methods. A typical range of MWCO for ultrafiltration (UF) membranes is 1–500 kDa⁴⁰, for nanofiltration (NF) membranes is typically below 1 kDa (0.15–0.3 kDa)⁴¹, for reverse osmosis (RO) is <0.2 kDa, and for “loose” NF membranes is 0.5–2 kDa³⁹. MWCO averages the pore size and does not consider molecular interactions (e.g., adsorption, charge interaction, solute–solvent interaction), which become relevant for separation in the range between UF and NF processes⁴². In comparison, retention by membrane bioreactor (MBR) can be more complicated as it depends on the formation and stability of a dynamic biofilm and since operational strategies and longer residence times can improve biodegradation of refractory organics, thus resulting in more uncertainty around allergen persistence⁴³. While all the food allergens in Table 1 are larger than UF, NF, Loose NF and RO pore sizes, best practices suggest targeting 30–50% of the pore size to contaminant size to get good rejection and the small size range of some of the allergens can bring them close to the recommended MWCO. Studies

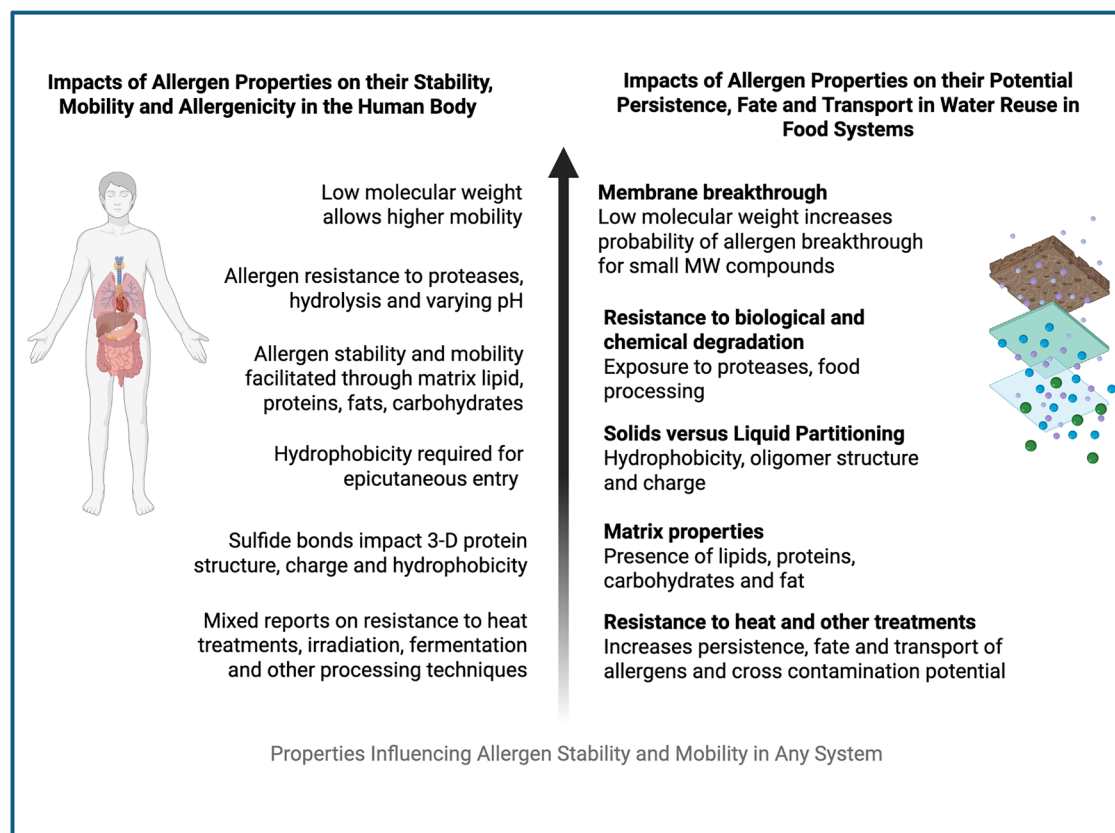


Fig. 2 | Properties of allergens that allow for their stability and persistence in biological systems and how these would impact fate and transport in water reuse systems. Created in BioRender. Maal-Bared, R. (2026).

have shown that, as a group, allergens tend to be more abundant and more stable than non-allergenic proteins from similar biological sources^{31,44}. These properties may increase the likelihood that allergenic proteins or immunologically relevant fragments persist following environmental or processing-related exposure. However, abundance and stability are not universal characteristics of all allergens and should be considered alongside other molecular and environmental factors when evaluating allergen persistence.

Other matrix properties also influence an allergen's susceptibility to enzymatic degradation. Allergen-lipid complexes may be formed with naturally occurring lipids, but can also be generated upon processing of foods and reuse of water in tanks⁴⁵. Most allergens contain surface areas with high proportions of hydrophobic regions. These hydrophobic regions can bind to lipids through electrostatic or lipophilic interactions, which also increases their effective size through complexation⁴⁶. The lipid content of food is known to prolong allergen mobility but the result of interactions with proteins and lipid in water systems is unknown. Allergen proteolysis is shown to differ between emulsions and solutions⁴⁷. Presence of other proteins, carbohydrates, and fats reduces absorption of allergens into the intestinal tract in some cases, but increases allergenicity in other cases (e.g., Brazil nut allergen *Ber e 1*)³². Overall, the presence of other compounds can impact disinfection efficacy through increased disinfectant demand during water conditioning. Also, allergens and non-allergens differed significantly in isoelectric point (pI), net charge, and electrostatic potential (EP), which influence stability and allergenicity. Singh et al. reported that, within their

dataset, allergens tended to have more acidic isoelectric points (pI; mean $\pm$ SE, 5.7 ± 0.15) and negative net charge, whereas non-allergenic proteins tended to have neutral-to-basic pI values (7.6 ± 0.16) and positive net charge⁴⁸. These physicochemical characteristics may influence the interactions of allergenic proteins with water, treatment surfaces, and other organic matter, potentially affecting their partitioning and adsorption during water treatment processes. Therefore, isoelectric point, as well as size exclusion, should be considered in the allergen removal treatment train that includes systems like RO. By removing whey protein fractions using UF upstream of RO, efficiencies of RO were improved in some studies⁴⁹.

Finally, treatment and processing methods can impact allergen stability and mobility. Milk allergenicity increases after homogenization and pasteurization, while fermentation and hydrolysis seem to result in milk allergen destruction⁴⁶. Extensive heating diminishes the allergenicity of egg white proteins, but limited evidence also suggests UV irradiation reduces the allergenic properties of eggs⁵⁰. Commercially available tree nuts are mostly heat-processed (dry and moist) or processed into butters, but many tree nut allergens show high resistance to temperature treatments. Seed storage proteins and lipid transfer protein family (e.g., Brazil nut and cashew nut) allergenicity was not impacted after roasting at 180°C for 15 min⁵¹. Recent work has shown that cashew protein⁵², crustacean protein in the form of tropomyosin⁵³ and gluten⁵⁴ all transfer to frying oil and persist at concentrations that result in cross-contamination of other products in cooking resulting in risk of an allergic reaction to consumers. Evidence regarding the effects of food processing on soy allergenicity remains limited and

Table 1 | Characterization of common food allergens and the limited information available on allergen removal from food processing wastewater

Allergen	Molecular Weight (kDa)	Length (amino acids)	Source	Isoelectric Point	Primary Amino Acid Sequence
6-lactoglobulin (Bos d 5)	18	162	Bovine (Dairy)	5.2–5.4	• High selectivity of α-LB and β-LG was achieved with a 30 kDa cellulose membrane with 90% recovery of β-LG in the retentate and 99% recovery of α-LB in the filtrate⁴⁷ • A 10 kDa UF membrane showed 100% recovery of lactose from the wastewater sample; the 10 kDa membrane showed 60–95% rejection of protein with high recovery achieved at TMP greater than 2.5 bar; lactose recovery improved by 5–20% for the 3 kDa membrane when temperature was increased above room temperature⁴⁸ • To treat isolated soy protein production wastewater for potential reuse of the recovered soy products, the 5 kDa membrane gave the best results resulting in the least reduction in permeate over time and 52% retention of protein⁴⁹ • No information is available on removal of Gliadin • Previous work showed that organic foulants with a molecular weight greater than 10 kDa contributed to flux decline and membrane fouling⁵⁰. These findings suggest that the lower molecular weight proteins in the feed water will be removed while the higher weight proteins for compounds like gliadin may contribute to MBR or RO fouling.
α-lactalbumin (Bos d 4)	14	123	Bovine (Dairy)	4.7–5.1	
Gly m 4	17	157	Soy	4.7–5.7	
ω Gliadin (Tri a 19)	32.6	280	Wheat (Gluten)	4.5–5.5	• No information is available on removal of Gliadin • Previous work showed that organic foulants with a molecular weight greater than 10 kDa contributed to flux decline and membrane fouling⁵⁰. These findings suggest that the lower molecular weight proteins in the feed water will be removed while the higher weight proteins for compounds like gliadin may contribute to MBR or RO fouling.
γ Gliadin ^a (Tri a 20)	37.1	327	Wheat (Gluten)	6.5–8.0	
α/6 Gliadin ^a	33	286	Wheat (Gluten)	7.5–9.0	
Non-specific lipid transfer proteins (nsLTP) ^b	10.1	102	Plant (Tree nuts)	0.7–9.2	• No information is available on the removal of nsLTP

All amino acid sequences retrieved on <https://www.uniprot.org/> and all proteins listed are hydrophobic.
^aSulfur-rich with disulfide bonds, while ω Gliadin is sulfur poor and known for celiac disease-triggering toxic epitopes.
^bHighly involved in lipid transfer and mobility.

inconsistent. Processing can alter the structure and immunoreactivity of soy proteins, with some treatments, including fermentation and hydrolysis, reported to reduce IgE-binding capacity, while other processing methods may have little effect and may not completely eliminate their allergenic potential⁴⁶. Overall, Verhoeckx et al.⁴⁶ concluded that the limited number of high-quality studies in clinically confirmed soy-allergic individuals prevents definitive conclusions regarding the effect of processing on soy allergenicity. In the case of wheat allergens, heating at high temperature in the presence of carbohydrates may induce the formation of protein aggregates with allergenic potential that are resistant to digestion and in order for enzymatic hydrolysis to reduce the allergenicity of wheat, specific proteolytic properties are needed⁵⁵. Thus, our understanding of how allergens behave in much more studied systems is still limited.

Indicator allergens and proteins

Based on this information and the data presented in Table 1 associated with four food allergens (i.e., dairy, wheat, soy and tree nuts), monitoring for allergens in water reused in food systems may fill a major knowledge gap. While allergen monitoring would be based on the foods and products being processed, using proteins like Gly m 1b due to its small molecular weight and non-specific lipid transfer proteins (nsLTP) due to its high resistance to degradation could be practical and most representative of water quality deterioration and cross-contamination with allergens. In addition, Table 1 lists the limited information available on allergen removal from food processing wastewater, which is intended to concentrate certain valuable food processing by-products for consumption (e.g., whey protein). There are no past studies on membrane size exclusion processes for the isolation of Gliadin or nsLTP from food processing wastewater, because their protein fractionation is not in high demand for protein products, such as those from bovine whey and soy whey. Table 2 includes a broader list of allergens and their general characteristics and what the implications may be for water treatment and reuse. The table also highlights potential allergens that may be used as indicators in water systems. Based on Table 2, proteins of interest that could act as indicators of allergen persistence in food processing water reuse systems include the whey protein Bos d 5, egg protein Gal d 1, peanut protein Ara h 2, tree nut protein Cor a 14, and sesame protein Ses i 1 due to their resistance to degradation. In addition some larger molecular weight proteins can serve as models for membrane retention or rejection, such as the whey protein Bos d 5, egg protein Gal d 3, soy protein Gly m 5, wheat protein Tri a 26, and tree nut protein Cor a 9.

Challenges and opportunities

As the food industry continues to optimize water reuse to reduce water consumption, the potential for higher allergen cross-contamination and risk to public health increases⁵⁶. There is a need to better understand the impacts of traditional and emerging food processing and water treatment technologies on protein structures and immunoreactivity (e.g., thermal, radiation, light, high-pressure, transient pressure, mechanical wave, plasma, electric field) and cross-contamination potential. Quantification of food allergens in food and other complex matrices remains challenging. Immunoassay-based methods, particularly enzyme-linked immunosorbent assays (ELISAs), are widely used for routine allergen analysis; however, their quantitative performance can be affected by matrix composition and processing history. Protein denaturation, aggregation, reduced solubility, epitope masking, non-specific interactions, and chemical modifications can reduce allergen extraction efficiency or alter antibody recognition, potentially resulting in inaccurate estimates of allergen concentration^{52,57}. ELISA, western blotting and immunochromatography all exhibit cross-reactivity and result in false positives⁵⁸. Although these

Table 2 | Approximate molecular characteristics of key allergens and their potential relevance to water treatment and reuse systems as indicators of treatment efficacy

Food source	Allergen	Major protein or family	Approx MW (kDa)	Approx length (aa)	Approx Isoelectric Point (pI)	Key molecular characteristic	Potential relevance to water persistence/treatment
Milk	Bos d 9	αS1-casein; casein	23.6	199–214	4.9–5.0	Highly abundant; phosphorylated; relatively acidic	Negative charge at typical process-water pH may influence adsorption; high abundance increases potential for cross-contamination
	Bos d 10	αS2-casein; casein	25.2	207	5.2–5.4	Phosphorylated casein; abundant	Similar potential charge/adsorption behavior
	Bos d 11	β-casein; casein	24.0	224	5.1–5.4	Abundant, amphiphilic phosphoprotein	Potential adsorption/aggregation and persistence
	Bos d 12	κ-casein; casein	19	190	~5.4–5.6	Glycosylated phosphoprotein	Surface interactions and aggregation may influence removal
	Bos d 5	β-lactoglobulin; lipocalin	18.3	162	5.1–5.3	Relatively stable globular protein; abundant whey protein	Good model for protein persistence and membrane rejection
	Bos d 4	α-lactalbumin; calcium-binding protein	14.2	123–142	4.5–4.8	Smaller globular whey protein	Useful size contrast with β-lactoglobulin
Egg	Gal d 1	Ovomucoid; protease inhibitor	28	186–210	4.0–4.6	Highly resistant to heat and proteolysis	Potential high persistence model; degradation resistance may support persistence
	Gal d 2	Ovalbumin; serpin-related protein	44	385	4.5	Most abundant egg-white protein; heat-sensitive relative to ovomucoid	High concentration but processing-dependent structural changes
	Gal d 3	Ovotransferrin	76–78	686–705	6.0	Large glycoprotein	Useful high-MW example for membrane retention/rejection
Soy	Gly m 4	PR-10 protein	17	157–158	4.7–5.7	Water-soluble; comparatively labile to heat/proteolysis	Potentially lower persistence despite allergenicity
	Gly m 5	β-conglycinin; 7S globulin	150–200	400–560	5–6	Multimeric storage protein; multiple subunits	High-MW/oligomeric model; potential membrane retention
	Gly m 6	Glycinin; 11S globulin	320–380	495	5.8–6.4	Large hexameric storage protein; multiple subunits	Strong size/structure contrast; likely important for membrane processes
Wheat	Tri a 19	ω-5 gliadin	65	439	7–7.3	Prolamin; structurally unusual, cysteine-rich; water insoluble	Persistence may be influenced by aggregation and protein-surface interactions
	Tri a 20	γ-gliadin	35–38	250–340	7.5–8.1	Prolamin; gluten-associated	Potential persistence/aggregation
	Tri a 26	High-molecular-weight glutenin	88	variable	6.0–8.1	Large polymer-forming gluten protein	Important high-MW/aggregation model
Peanut	Ara h 1	7S vicilin-like globulin	63–70	619	5.5–8.3	Abundant major allergen; oligomeric	Critical model for high-MW allergen retention/persistence
	Ara h 2	2S albumin	16.6–19.5	172	5.2	Highly allergenic; relatively small, disulfide-rich	Critical small/stable protein model
	Ara h 3	11S globulin	14–45	507–538	5.5–5.7	Storage protein; processed into subunits	Size and oligomerization may influence membrane removal
Tree nuts	Ara h 6	2S albumin	15	~124–145	5.2	Disulfide-rich storage protein; structurally related to Ara h 2	Useful comparison with Ara h 2
	Cor a 9	11S globulin; legumin	59	514–515	6.46	Major hazelnut storage allergen	High-MW storage protein; useful for comparison with Ara h 3/Gly m 6
	Cor a 14	2S albumin	14–16	147	8.22	Stable seed-storage protein	Small, disulfide-rich allergen potentially resistant to degradation
Fish	Gad p 1	β-parvalbumin	~11.5–12	109	4.0–4.5	Highly stable calcium-binding protein	Critical small/stable allergen model; potentially persistent despite low MW
Crustacean shellfish	Pen m 1	Tropomyosin	38	284	4.5	Heat-stable, highly conserved muscle protein	Critical cross-reactivity/persistence model
Sesame	Ses i 1	2S albumin	9	153	6–7.3	Seed-storage protein; disulfide-rich	Useful small, relatively stable allergen
	Ses i 3	7S globulin/vicilin	45	562–585	5.0–5.5	Storage protein	Size/oligomerization contrast with Ses i 1

challenges have been well documented in food matrices, considerably less is known about the quantitative recovery and immunoreactivity of allergens in food-processing waters, where low concentrations, dissolved organic matter, surfactants, salts, and other treatment-related constituents may introduce additional matrix effects. PCR is an option, but experiences low sensitivity for some heat-processed foods due to extraction of an inadequate amount of DNA for testing⁵⁹. Mass spectrometry methods increasingly offer a promising alternative technique to overcome some of these issues, as protein conformations are not required for detection and harsh (reducing/denaturing) extraction conditions can be used more effectively⁵². LC-MS/MS Method validations are targeting multiple targets with recent work targeting seven allergens simultaneously⁶⁰. More investigation is necessary to confirm if and how proteomics could be used for the detection of target food allergens and produce results in a timely manner that is protective of public health. Previous work using shotgun proteomics in wastewater treatment plant sludges mainly detected extracellular proteins involved in binding, catalysis and degradation⁶¹, but it is unclear if allergens in wastewater or conditioning water could be detected by proteomics. In addition, effective, sensor-based, extraction-free water quality monitoring parameters and controls (e.g., TOC, turbidity)^{62,63} that can be used as allergen surrogates in recycled water and permeate are needed. Fluorescence and absorbance spectroscopy (e.g., UV₂₅₄) can serve as a sensitive indicator of water quality changes and dissolved organic material in certain water matrices^{64,65}. Monitoring transmembrane pressure (TMP) differential is commonly used to detect membrane fouling⁸ and may also be helpful in identifying allergen breakthrough and persistence in the water-food nexus. More data is also needed to correlate occurrence of food allergens and cross-contamination events with online water quality monitoring parameters to develop appropriate ML/AI algorithms to predict allergen removal and risk to consumers². In addition, the inclusion conditioned water properties in food industry blockchains could aid detection accuracy, transparency, and addressing allergen cross-contamination in food traceability ultimately leading to improved public health protection⁶⁶.

Conclusions

- The food industry is increasingly reusing water and treated wastewater effluent to support critical growing, harvesting and processing functions with water treatment targets being tailored to the intended end-use.
- Food allergies are an important public health problem that impacts 1–10% of the global population and their frequency and severity is reported to have increased over recent decades.
- Research showed that 92% of companies produced food products containing one or more of the nine major food allergens and despite 96% of companies having detailed food allergen control plans in place in alignment with best practices, nearly half had at least one food allergen-related recall within the past 5 years.
- Water treatment and conditioning for reuse in the food industry is a common practice, but treatment trains, disinfectant dosing and monitoring systems are not standardized.
- To our knowledge, there has been no research on the occurrence of allergens in food systems processing water, and the potential for allergen cross-contamination has not been explored.
- The same properties that allow stability and persistence of allergens in biological systems can also potentially result in persistence and impact fate and transport in water reuse system in the food industry.

- Given the large number of food allergens, the use of certain allergens as indicators can increase detection success and provide critical data to improve water reuse practices in the water-food nexus.
- Allergens of interest that could act as indicators of allergen persistence in food processing water reuse systems include the whey protein Bos d 5, egg protein Gal d 1, peanut protein Ara h 2, tree nut protein Cor a 14, and sesame protein Ses i 1 due to their resistance to degradation. In addition, some larger molecular weight proteins can serve as models for membrane retention or rejection, such as the whey protein Bos d 5, egg protein Gal d 3, soy protein Gly m 5, wheat protein Tri a 26, and tree nut protein Cor a 9.
- Improved online, sensor-based controls and laboratory analytical methods are needed to improve detection of allergens in unusual matrices and facilitate use of data for ML/AI and blockchains.
- There is a need to better understand persistence of allergens in conditioned water in food systems and the impacts of traditional and emerging food processing and water treatment technologies on protein structures and immunoreactivity and cross-contamination potential.

Data availability

No datasets were generated or analysed during the current study.

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R.M.B.—Conceptualization, Writing Original Draft, Writing Reviews and Edits, Visualization, Supervision, Resources; G.E.—Writing Original Draft, Writing Reviews and Edits; I.W.—Conceptualization, Writing Reviews and Edits, Administration; J.W.—Writing Reviews and Edits;

J.C. — Writing Reviews and Edits; D.W. — Writing Reviews and Edits; C.L. — Conceptualization, Writing Reviews and Edits, Supervision, Administration.

Competing interests

R.M.-B., G.E., I.W., and C.L. have no competing financial interests to declare. J.W., J.C., and D.W. work for Hygiena, which produces Health Diagnostics solutions that promote food safety throughout the value chain. This includes allergen testing solutions. They also have no competing financial interests to declare. There are no pending patents related to this work. None of the authors act as editor-in-chief, associate editor, or guest editor for npj journals. Competing interest statements have been included both in the manuscript and in the submission system for full transparency.

Additional information

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